

Business Problem

We have a retail store which depends on the sales of its products to pool of its customers from the USA. The main purpose is to inspect how the store is doing in terms of sales and profit. It can be assumed that there are no other income streams of the company. The main source of revenue is the sales of its products of three categories and seventeen subcategories. This means the revenue is synonymous as sales.

Some of the areas which have been focused in this analysis is the total sales and total revenue, best performing products and customers to determine valuable costumers and top and bottom products to remedy with specific discounts and sales tactics, and to ascertain the best performing regions to focus on them.

Dataset

As we have considered a hypothetical retail store, therefore we have just accessed an open resource sample data for a superstore. There are total 21 columns in the dataset. Each column is an attribute related to the sales of the superstore. For instance, order ID is the unique ID for each order/sale for the superstore.

Methodology and Tools

The methodology adopted here is quite similar to that of a normal data analytics project. First of all, we tried to understand the data to relate it to the business problem we are dealing with. Then we applied some data wrangling techniques and we further try to dig deep into it by doing the Exploratory data analysis. After that we built a dashboard and try to come up with the recommendations and business solutions.

We used a mix of tools for this. The basic analysis has been done in Python, the power BI has been used to built the dashboard and some problems has been tackled using the SQL.

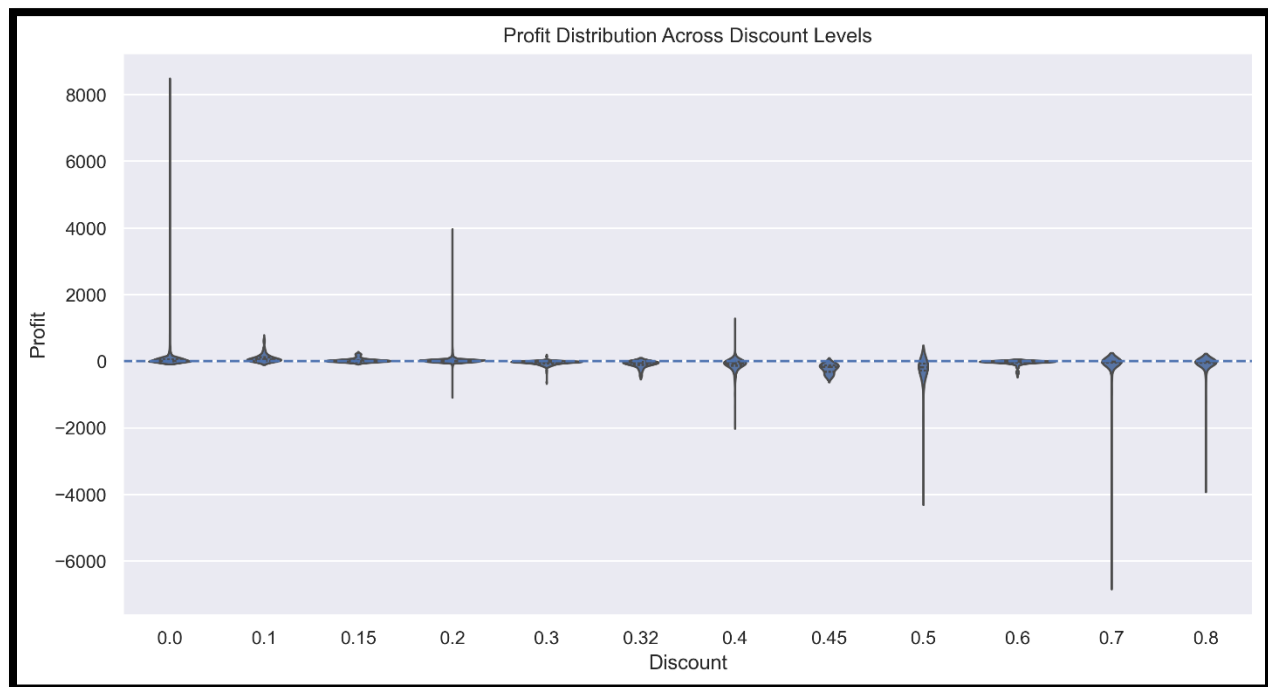
Analysis

Data Understanding

The dataset contains 10194 rows and 21 columns/variables. Row ID (as obvious), sales, discount, and profit are numerical (int and float). Order date and ship date are date fields, whereas segment, category, and subcategory along with some other are categorical variables.

Data Wrangling

There are no missing values within our dataset. Moreover, no presence of duplicates has been determined. The data is showing a very high negative profit below the mean which is suspicious. There seems to be no invalid values. There seems to be the presence of outliers through IQR method. The negative side outliers can be justified. In this case the most plausible reason could be the heavy discounts on very expensive items. Some expensive one-off items (a \$6,600 loss on a Cubify 3D printer, \$3,800 on another one, \$3,700 on an electric binding system). These are high-ticket equipment where either a huge discount or an unusual return/pricing situation wiped out the entire sale. The relationship of discount and profit has been shown by a violin plot.

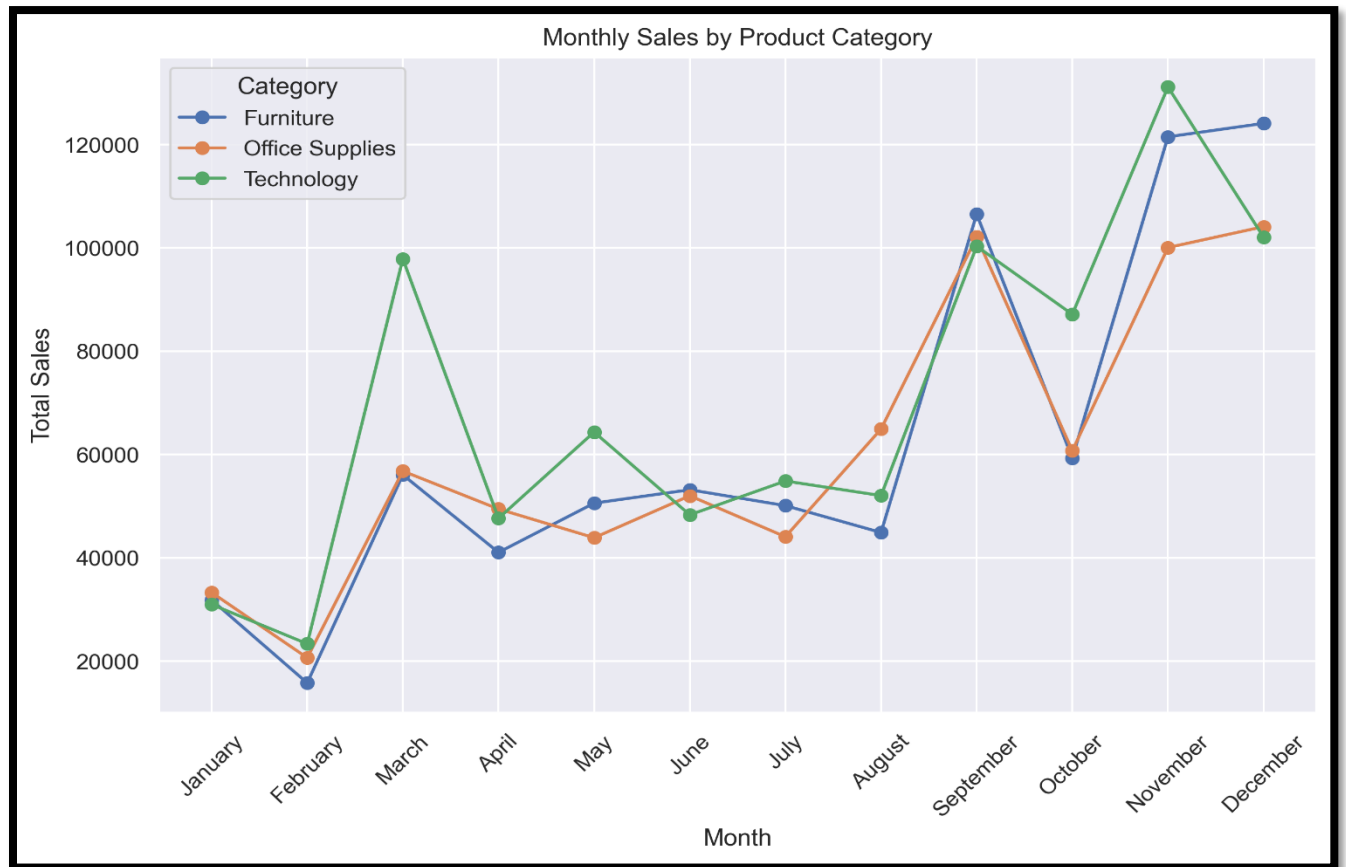


The above graph shows that a maximum of 20% of discount should be given as it is highly positively correlated to the profit. Discounts above 20% generated negative profits.

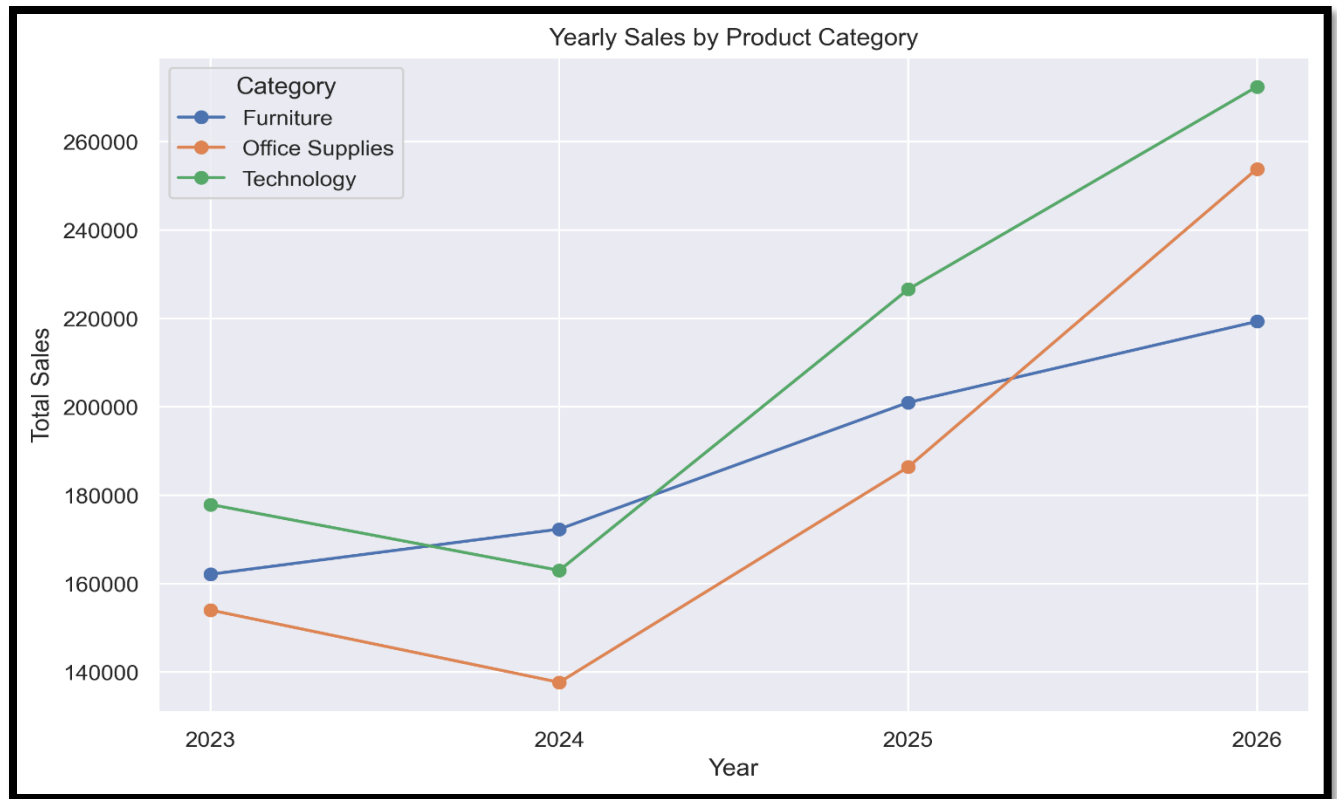
This also suggest that the outliers in this case are plausible so we don't need to remove them.

Exploratory Data Analysis

There seem to be seasonal patterns as sales high in certain months like in March and September/October. If we see monthly sales for each category, we can see that technology products outperform the other two.

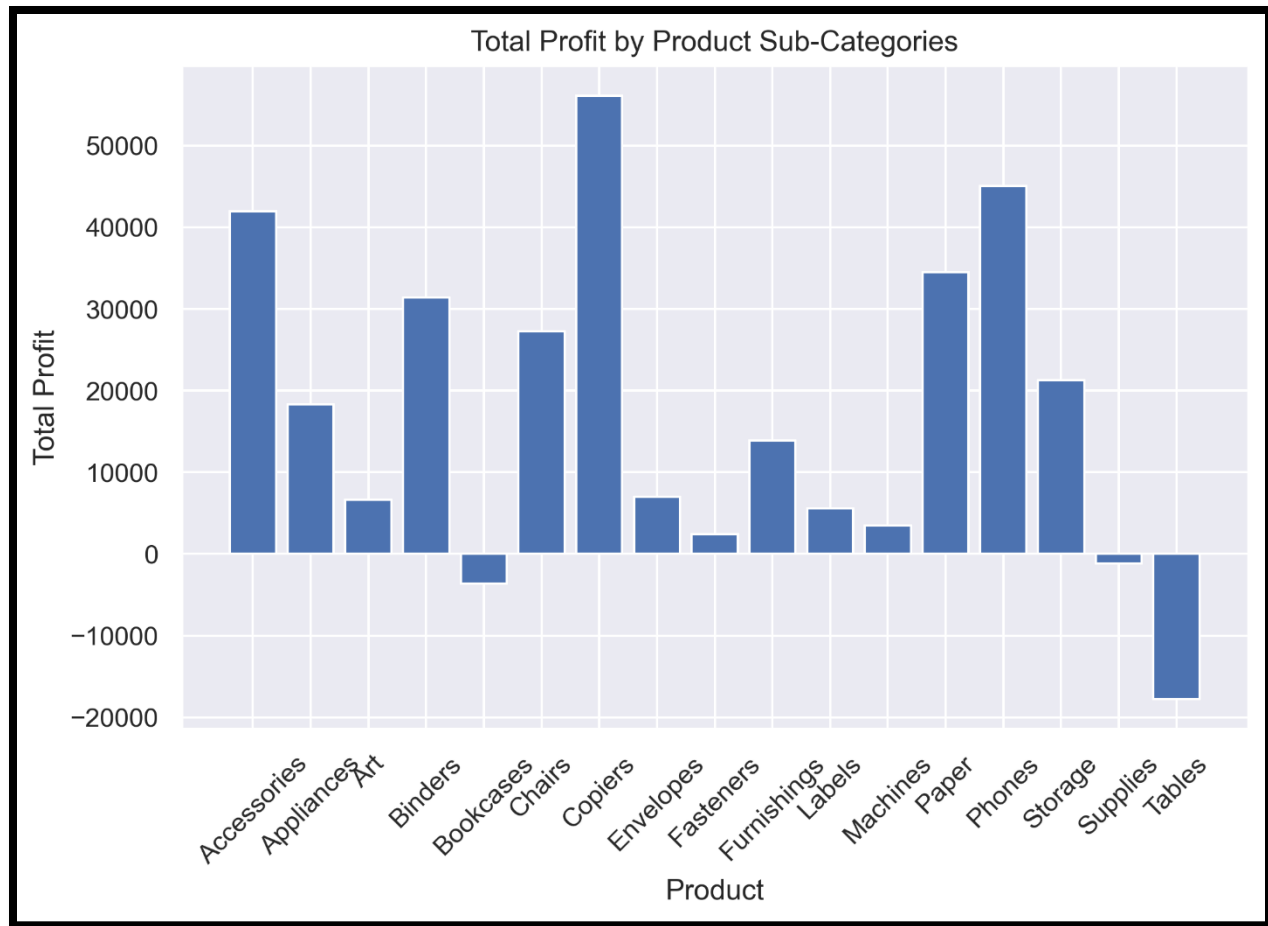


The yearly sales generally show an increasing trend with again the technology category is showing the best sales. The reason again could not be the quantity itself but the prices of very expensive tech products.



The top 2 regions in terms of profit are West (1st) and East (2nd). Both regions are generating around 100k profit. East and West win on a combination of higher sales volume, lower average discounting, better profit margins, and a couple of dominant high-revenue states (California, New York) anchoring each region. At the other end Central's heavy discounting habit is the clearest reason it underperforms relative to its sales size.

The top 10 costumers in terms of sales and profit are different as some of the high sales customers like Sean and Hunter may generate highest sales but not profits. So, the top customers will obviously be based on the profit. Similar is the case with products. The bottom 10 products are all generating negative profits. It can also be noted that 90% of these products carry a 20 – 53% discount (the range yielding negative profits).



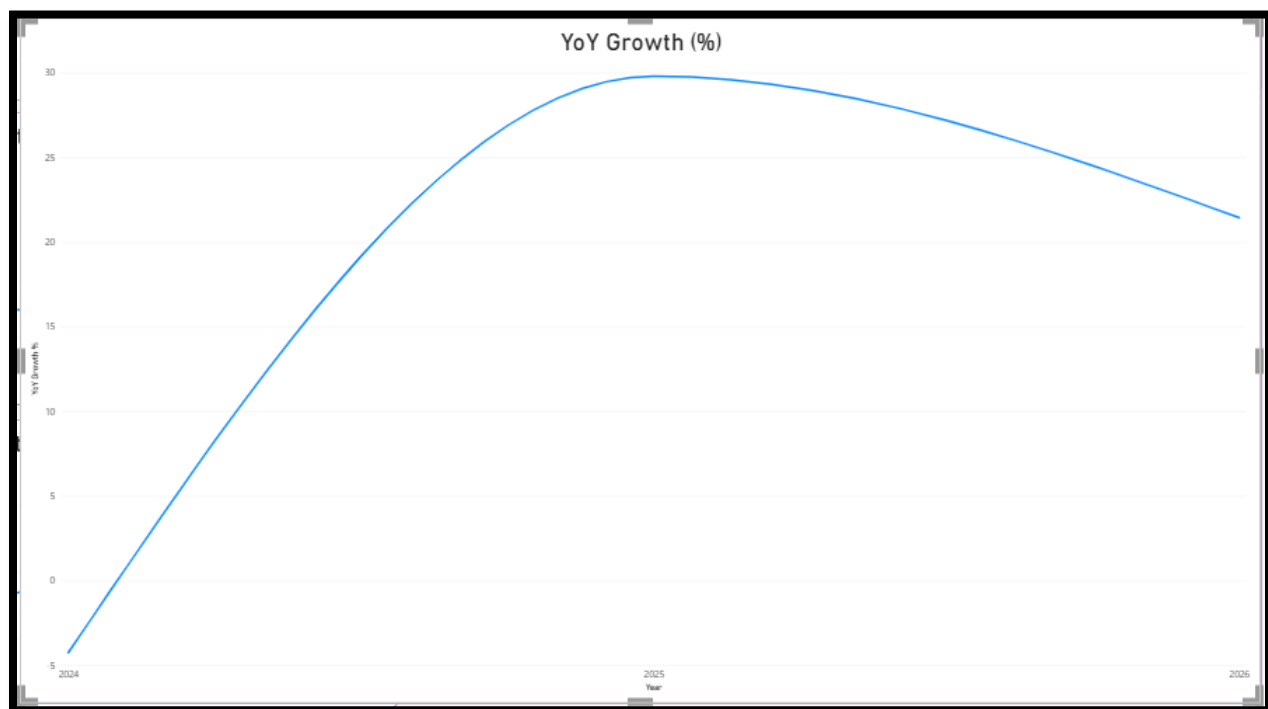
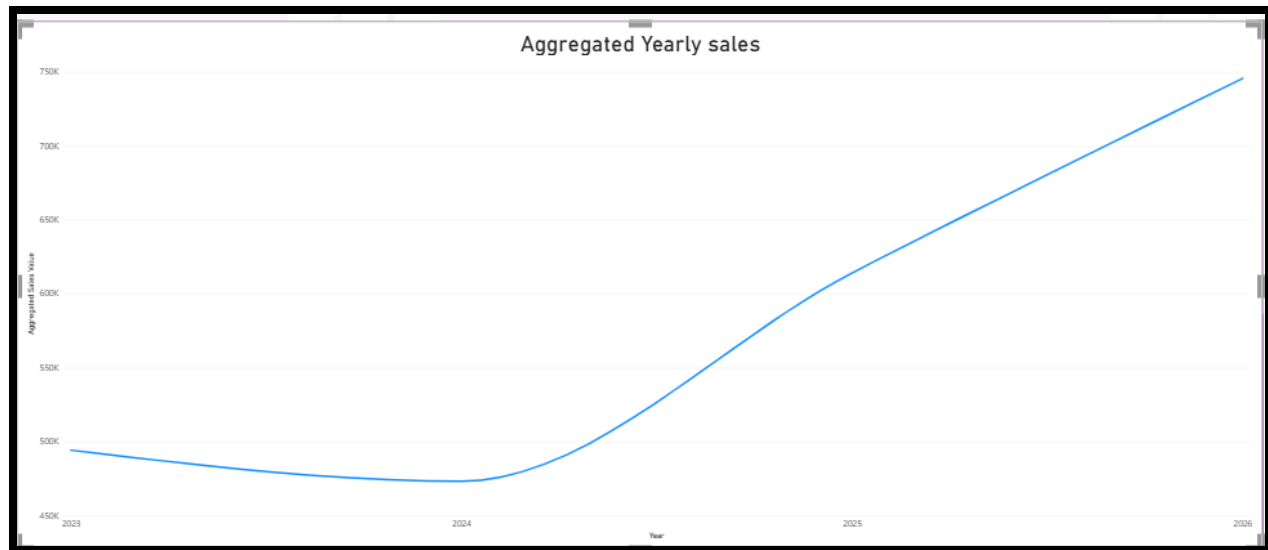
Out of the total 17 product sub-categories, three (bookcases, supplies, and tables) are producing losses. They fall under the category of furniture and office supplies.

Dashboard

The dashboard usually gives the same information presented in the EDA about the data. Only the yearly sales and yearly growth rates gives additional information about the yearly trend.

We can see there is the decline in yearly sales which indicates negative growth rate for the year 2024. The growth rate rises significantly in 2025 to 30% then it declines to 21% in 2026.

Here we can assume that the company has been established in 2023. Thus, it is obvious for any business that it runs in losses in initial year as it has to bear the sunk costs including in the fixed costs. Then we can see there is a very sharp rise in 2025 which can be rapid gain in customers, promotion, discount campaign and other micro and macroeconomic factors.



Key Insights

The company is doing quite well which can be tapped through positive growth rates and substantial sales and profits. As we can see the retail market is highly saturated and potential competitors are present in a US market so it would be very challenging to capture a good market share considering market giants like Staples, Office Depot (The ODP Corporation), and W.B.

Mason. As our retail company is a newbie in market. We can observe that the company is focusing on B2C model instead of B2B model.

The total orders are above ten thousand but not very considerable. The profit margin is around 13% with a great potential for improvement. The aggregated monthly sales and profits are showing an increasing trend as well as seasonal patterns with peaks in March, September, and November. As expected, the high sales regions are East and West.

The technology category products are leading in terms of sales but we should also notice that the other two categories are almost at par with the leading category (furniture being second and office supplies being third). In terms of profit, furniture is producing very less profit indicating either the Cost of Goods sold is almost equal to the revenue or showing very high maximum (70%) and average (17%) discounts on it. The profit margin on it is also lowest as expected. Among the subcategories some are considerable leading. For instance, chairs and phones are the top two leading subcategories. The top ten profitable products fall under the category of technology and office supply, whereas the bottom 10 products are a mix of all the three categories. The West region shows an interesting result as the top category is furniture.

The category, region, and year wise results show a little bit variation than general results.

Top six customers with respect to sales and profit are same. The remaining 4 profitable customers are not generating the top sales. Also, none of the most profitable customers are having highest number of orders.

In terms of profit, West and East are also doing the best. In terms of profit margins, all regions fall in the range of 5 – 15% which is considered typical for retail industry. As central has the lowest profit margin so the discount patterns also show that the maximum (80%) and the average (24%) discount is being offered in this region. The quarterly sales also show an increasing trend with Q4 outperforming than the rest.

Recommendations

First of all, let's focus on the regions. As West and East are already well catered and it also shows similar patterns in competitors. So, to be differentiated among the competitors, the retail business should focus on the other two regions. Firstly, they should revisit their discounting strategy and should keep their discount in the range of 10 – 20%. The 20% discount is ideal as it is directly related to extremely high profits.

Secondly, the company should also lessen the discount on the furniture and review the pricing of it. They can introduce better quality highly priced furniture.

The tables, bookcases, and supplies are generating losses. Either the related products can be discontinued or the discount should be removed from it. More depth analysis by looking at their COGS should be made.

The profit margin on the top 10 products should be increased by advertising it more often. In this case, it was observed that the most frequent buyers don't produce the highest sales and profit. Therefore, some tactics like loyalty programs and some special discounts on top products should be offered so the loyal costumers start to buy them.

Generally, the retail store should increase its product line as well as product categories to tap to the demand of large audience in less demanding regions of Central and South. For instance, if currently the retail business in focusing on office furniture, then it should also include home furniture as well. It can also tap in the category of clothing and shoes by including office wear clothing and shoes. This is a long-term recommendation which it should test first.

Another suggestion would be to introduce some unique items like hand made furniture by buying from the less privileged markets particularly in the South Asia, Africa, and South America. These products may have high profit margin potentials